

Arden's Lemma

Let Q, P be reg. lang. s.t. $\epsilon \notin P$. Then, $\exists$ a unique solⁿ to $R = Q + RP$ as $R = QP^*$.

Proof: ($\Rightarrow$) By induction, $QP^j \subseteq R$. $Q \subseteq Q + RP = R$.

$$QP^{j+1} = QP^j \cdot P \subseteq RP \subseteq Q + RP = R$$

($\Leftarrow$) Induct on word length. let the smallest be a . Then $x \in R \Rightarrow x \in Q$ or $x \in RP$, can't be P as $\epsilon \notin P$. Hence $x \in Q \Rightarrow x \in QP^*$.

Again $x \in RP \subseteq Q$; If Q done, else $\exists y \in R, z \in P$ s.t. $x = yz$; $y \in QP^*$; $|y| < |x|$
Hence $yz \in QP^* \Rightarrow x \in QP^*$. □

Exⁿ: If $\epsilon \in P$, then R can be any of $\{AP^* \mid A \geq Q\}$.

Application: DFA $\rightarrow$ Reg Ex:



Accepts $L = \{s \in \{0,1\}^* : \text{sho an even no. of strings}\}$

$$A = \epsilon + Ab + Ba$$

$$B = Aa + Bb$$

$$\Rightarrow B = Aab^*$$

$$\text{so, } A = \epsilon + Ab + Aab^*a$$

$$\Rightarrow A = \epsilon + Ab + Aa(b)a^*$$

$$\Rightarrow A = \epsilon + A(b + aba^*)$$

$$\Rightarrow A = \epsilon(b + aba^*)^*$$

$$\text{Hence } A = (b + aba^*)^*$$